

Linear Regression : Assumptions

Programming for Analytics (Python)

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A Quick Analogy

- If the question bank is provided , you can have a high exam score (high R^2) by memorizing the answers
but
- if the exam questions change (new data), you'll fail because you didn't actually understand the subject (violated assumptions).

Linearity

- **What it means:**
The relationship between the independent variables (X) and the dependent variable (Y) should be **straight-line (linear)**.
- **Example (Business):**
If you're predicting **sales** based on **advertising spend**, sales should increase proportionally as ad spend increases.
If doubling ad spend doesn't consistently increase sales (maybe it levels off after a point), the relationship isn't linear.
- **Why it matters:**
If the true relationship is curved, the linear model will give **biased predictions**.

Mean of Residuals = 0

- **What it means:**

On average, your model's prediction errors (residuals) should **cancel out** — some positive, some negative, but mean ≈ 0 .

- **Example (Business):**

If your model predicts **monthly revenue**, sometimes you'll overestimate and sometimes underestimate, but overall the errors should balance out.

- **Why it matters:**

A non-zero mean suggests the model is **systematically under- or over-predicting**.

Homoscedasticity (Constant Variance of Errors)

- **What it means:**
The residuals should have **constant spread** (variance) across all levels of X.
- **Example (Business):**
When predicting **employee salaries** from **experience**, the prediction errors should be roughly the same for junior and senior employees.
If senior employees' salaries vary wildly but juniors' don't → that's **heteroscedasticity**.
- **Why it matters:**
Unequal variance makes your **standard errors and confidence intervals unreliable**.

Normality of Error Terms

- **What it means:**

The residuals (errors) should follow a **normal distribution** — bell-shaped and symmetric around zero.

- **Example (Business):**

When forecasting **monthly product demand**, the small errors should be most common, and large over/under predictions should be rare.

- **Why it matters:**

Normal errors allow you to use **accurate significance tests** (t-tests, F-tests).
If residuals are skewed, your p-values may be wrong.

No Autocorrelation of Residuals

- **What it means:**
Errors should be **independent** of each other — one error shouldn't predict the next.
- **Example (Business):**
In **time series sales data**, if your model consistently underpredicts sales for several months in a row, the errors are correlated → **autocorrelation**.
- **Why it matters:**
Autocorrelation means your model is missing a pattern (like seasonality or trend), leading to **biased predictions**.

No Perfect Multicollinearity

- **What it means:**
Independent variables shouldn't be **perfectly or highly correlated** with each other.
- **Example (Business):**
If you include both “**marketing spend**” and “**total advertising budget**” in your model, they're likely measuring the same thing → **multicollinearity**.
- **Why it matters:**
It becomes hard for the model to know which variable actually influences Y, leading to **unstable coefficients** and confusing interpretations.

Fixing Assumption Violations

Assumption	If Violated	Fix / Solution
Linearity	Model misses pattern	Transform variables, add polynomials, or use nonlinear models
Mean of residuals = 0	Model biased	Include intercept, add missing predictors
Homoscedasticity	Unequal error spread	Transform Y, WLS, or robust SE
Normality	Skewed errors	Transform Y, remove outliers
No autocorrelation	Time-related errors	Add lag variables, use ARIMA
No multicollinearity	Redundant predictors	Drop, combine, or use PCA

What about neural networks ?

Assumption	Needed in Neural Networks?	Comment
Linearity	✗	NNs learn non-linear patterns automatically
Normality of errors	✗	NNs focus on prediction, not statistical inference
Homoscedasticity	✗	NNs can handle variable error variance
No autocorrelation	⚙️	Not needed; special architectures handle dependence
No multicollinearity	⚠️	Doesn't break model, but can cause inefficiency
Mean of residuals = 0	✓	Usually achieved by training

Let's Look at the code
